

Omega v1.0

(Confidential Document)

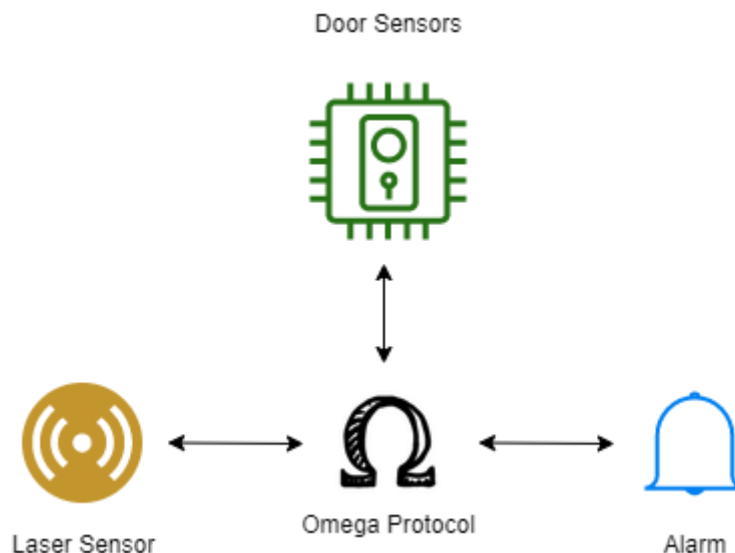
Use Case

Omega is a proprietary protocol that is used to strengthen prison security and maintain them as a Smart Penitentiary Facility.

To avoid problems such as possible prison breaks and rival gang member fights, we have fixed different sensors including positioning devices, video cameras and perimeter monitors. If something is detected by the sensor then the video cameras will turn to a given direction, and the main gate will be closed.

All these devices work with `omega` protocol, which is efficient and can be used flexibly across devices. Each prisoner is closely monitored with video cameras, apart from a 10 minute maintenance period during the day time. As this is critical period, we request the guards to move out of the base and activate the laser systems (E1,E2 & E3). Any failure in laser systems will trigger the alarms (A1, A2, A3 & A4) thereby alerting the guards, who will move to protect all prison entrances.

Architecture



Omega is a RF protocol. This protocol can be used in devices that operate at 1.8-5.5V, with a frequency band in the range of 433.92 MHz. We have used an in-house Omega prototype to convert RF signals to message format and vice versa.

Alarm Packet Structure

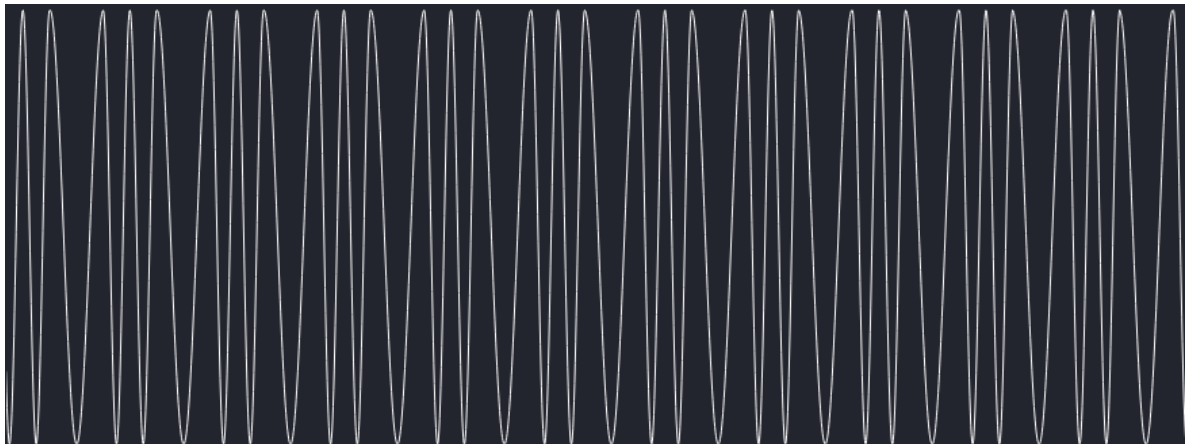
Packet Type	Packet Identifier
Suppress	0xF1
Alert	0xA0
Broadcast	0x03

Broadcast: In order to consume low power, all the alarms are in idle mode at regular intervals. All sensors broadcast messages to nearby devices with an allowed 0.05 seconds delay.

Alert: If broadcast mode is disrupted, then the nearby receiver (which acts as master) sends a signal to all alarms to put them into alert mode.

Suppress: This mode helps in silencing the alarms. This works by sending a suppress packet to the respective alarm.

Laser Packet Structure



Packet Type	Packet Identifier
Move left	0x91
Move right	0x53
Turn off	0xFF
Turn on	0xEE

To avoid packet replay and injection attacks we've implemented cyclic redundancy check validation, referring to microchip solutions port for STM32 with initCRC(0x1d0f). Invalid packets get discarded at the transceiver end.